

## Practical problems with equations



1 2

2

a  $x + y = 13$

b 8 cm

3 3500 people

4  $x = 18$

5 18

6 48 hours

7  $\frac{36}{5}$  hours

8

a 2

b 9

9 10 minutes

10 4

11

a 8 mph

b 16 mph

12

a

|            | Distance (miles) | Rate    | Time               |
|------------|------------------|---------|--------------------|
| Upstream   | 15               | $n - 3$ | $\frac{15}{n - 3}$ |
| Downstream | 30               | $n + 3$ | $\frac{30}{n + 3}$ |

b  $n = 9$

13  $n = 81$



14

a 45 mph

b 66 mph

15 23 mph

16

a  $(1 - k)t$

b

i  $d = 380kt$

ii  $d = 350(1 - k)t$

c  $k = \frac{35}{73}$

d 1184 miles

17

a  $x = 7$

b  $x = 3$

c  $x = 2$

18

a  $m + 71 = 403$

b  $m = 332$

19

a  $p = 164$

b \$436

20

a  $x + 2$

b  $x + 4$

c  $2x + 4$

21 10, 12, 14

22 13, 15, 17, 19

23

a  $\frac{1}{4}x = 3x - 22$

b  $x = 8$

24

a  $189 = 21x + 105$

b  $x = 4$

25

a  $24x + 3$

b  $x = 13$

26

a  $2x - 3$

b 9 inches



c 15 inches

27 14, 15, 16

28

a  $x = \frac{74}{9}$

b  $\frac{296}{9}$

29  $x = 6$

30

a  $x = -1$

b  $8^\circ$

31  $x = 12$

32  $x = 5$

33

a  $x = y + 1.36$

b  $8x + 9y = 119$

c  $y = 6.36$

d  $x = 7.72$

34

a  $x = y - 4$

b  $2(x - 4) + 2y = 12$

c  $y = 5$

d  $x = 1$

35

a  $x = 7y$

b  $x + 2 = 5(y + 2)$

c  $y = 4$

d  $x = 28$

36  $11^\circ\text{C}$

37

a 4

b 60

38

a \$271.12

b Underestimate

c \$1.73

d 32 years

e 2030

39  $B = \$14\,736.84$



40 17 years old

41 36 000 kg

42 28 advertisements

43 7 nights

44 240 L

45 10 years old

46  
a 8

b 81

47 4 hours

48  $2w^2 + 13w = 260$

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## Additional questions

49

a  $x = 2$

b  $m = -3$

c  $p = 4$

d  $k = -9$

50

a  $m = 7$

b  $k = 5$

c  $t = 1$

d  $d = -7$

e  $f = 4$

f  $r = -1$

g  $w = -3$

h  $p = -5$

51

a  $x = 8$

b  $x = 20$

c  $x = 75$

d No solution

52

a  $h = 4$

b  $x = 4$

c  $x = 7$

d  $m = 2$

e  $x = -3$

f  $p = -2$

g  $k = 12$

h  $x = 20$

i  $x = 16$

j  $x = 11$



$x = 4$

l  $y = -3$

m  $x = 70$

53

a  $x = -5$

b  $x = \frac{2}{5}$

c  $x = -\frac{22}{19}$

d  $x = \frac{11}{4}$

e  $k = 8$

f  $y = 8$

g  $x = 22$

54

a One solution

b Infinitely many solutions

c Infinitely many solutions

55  $x = 3$

56  $x = 4$

57  $x = 16$

58  $x = 13$  inches

59  $p = \$186$

60  $k = 14$

61

a Let  $x$  be the exact number of people at the concert. Stormie's guess was 800 too many, so Stormie's guess was  $x + 800$ . Pierce guessed 300 too few, so Pierce's guess was  $x - 300$ . The average of their guesses is 4250, so that means

$\frac{\text{Stormie's guess} + \text{Pierce's guess}}{2} = 4250$ . We can then substitute in the equations we have found for Stormie's and Pierce's guesses to get the equation  $\frac{x + 800 + x - 300}{2} = 4250$ .

b 4000

62

a  $12 = 0.5 \cdot L$

Where  $L$  is the number of laps.

b  $L = 24$

c Assuming both of them used their own cart, they would need to pay \$2 as a deposit. So



63 Bassam should have multiplied by  $\frac{1 \text{ mi}}{1.6 \text{ km}}$  to convert from km to miles.

64

- a True. If you substitute the solution into the unknown variable and calculate each side separately, you should end up with the same value on both sides.
- b False. For example, the equation  $x + 1 = 3$  has the solution  $x = 2$ , and the equation  $x - 1 = 1$  also has the solution  $x = 2$ .
- c False. If the variable terms do not cancel out, then the equation must mathematically have one solution.
- d True. For example, if a value of  $x$  satisfies  $2x + 3 = 1$ , then it will also satisfy  $4x + 6 = 2$ .

65

- a
  - i  $3n + 3 = 4n + 6$
  - ii One solution
- b
  - i  $2n + 6 = 2n + 5$
  - ii No solution
- c
  - i  $4n + 2 = 4n + 2$
  - ii Infinitely many solutions

66  $y = 8$

67  $B = \$15\,652.17$

68  $y = 16$  years old

69  $x = 12$

70

- a  $x = 17$
- b 17, 18, 19
- c No, because the equation used to solve for  $x$  in part (a) has only one unknown,  $x$ , to the power of one. Therefore there is only one solution.